

Lab 3: Component Modelling & Architectural Pattern Selection

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Lab	Lab 3 – Component Modelling & Architectural Pattern Selection
Scenario used	Problem Statement #18: Clinical Trial Subject Recruitment Engine (as per Lab 1 problem-statement PDF, used here in place of the sample Coffee Kiosk scenario)

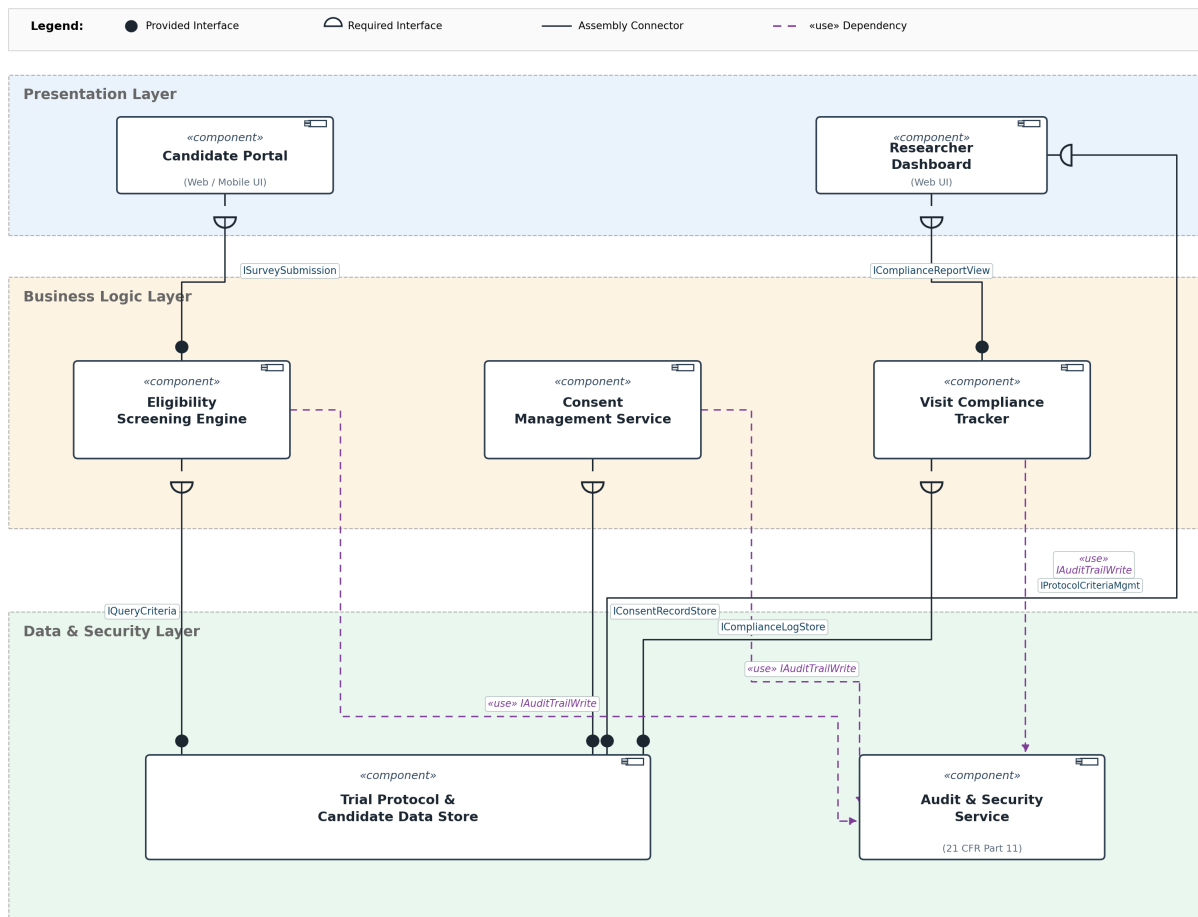
Scenario Note

This submission applies the Lab 3 component-modelling and architecture-selection workflow to the system described in Problem Statement #18 (Lab 1): a Clinical Trial Subject Recruitment Engine for medical research institutions. The platform performs eligibility pre-screening of trial candidates against protocol inclusion/exclusion criteria, tracks informed-consent milestones, and logs visit compliance, while meeting 21 CFR Part 11 electronic-record and audit-trail requirements. The two stakeholders are the Trial Candidate and the Clinical Researcher.

1. Component Diagram

Architectural style: Layered Architecture (Presentation → Business Logic → Data & Security).

UML Component Diagram — Clinical Trial Subject Recruitment Engine (Layered Architecture)



Components (7)

- Candidate Portal – web/mobile UI where candidates submit screening survey responses (Presentation Layer).
- Researcher Dashboard – web UI for researchers to manage protocol criteria and review compliance (Presentation Layer).
- Eligibility Screening Engine – evaluates candidate responses against inclusion/exclusion criteria (Business Layer).
- Consent Management Service – tracks informed-consent milestones (Business Layer).
- Visit Compliance Tracker – logs and monitors subject visit compliance (Business Layer).
- Trial Protocol & Candidate Data Store – persists protocol criteria, candidate, consent and compliance records (Data & Security Layer).
- Audit & Security Service – generates cryptographic audit trails for 21 CFR Part 11 compliance (Data & Security Layer).

Interfaces (7 – exceeds the minimum of 4)

- `ISurveySubmission` – Candidate Portal (required) ↔ Eligibility Screening Engine (provided).
- `IComplianceReportView` – Researcher Dashboard (required) ↔ Visit Compliance Tracker (provided).
- `IProtocolCriteriaMgmt` – Researcher Dashboard (required) ↔ Trial Protocol & Candidate Data Store (provided).

- IQueryCriteria – Eligibility Screening Engine (required) ↔ Trial Protocol & Candidate Data Store (provided).
- IConsentRecordStore – Consent Management Service (required) ↔ Trial Protocol & Candidate Data Store (provided).
- IComplianceLogStore – Visit Compliance Tracker (required) ↔ Trial Protocol & Candidate Data Store (provided).
- «use» IAuditTrailWrite – usage dependency from Eligibility Screening Engine, Consent Management Service, and Visit Compliance Tracker to the Audit & Security Service.

2. Written Justification

We chose Layered Architecture for the Clinical Trial Subject Recruitment Engine.

Architectural Choice

The system is organised into three horizontal layers: a Presentation Layer (Candidate Portal, Researcher Dashboard), a Business Logic Layer (Eligibility Screening Engine, Consent Management Service, Visit Compliance Tracker), and a Data & Security Layer (Trial Protocol & Candidate Data Store, Audit & Security Service).

Two Reasons

- Regulatory workflow fit: candidate- and researcher-facing interfaces, protocol/eligibility/consent business rules, and record-keeping are naturally distinct concerns in a clinical-trial platform. Separating them into layers lets each concern evolve (e.g., updated trial protocols, new consent forms) without touching the UI or the storage/audit mechanisms.
- Consistent compliance logic: centralising trial-protocol data and the audit trail in one lower layer lets every business-layer service (eligibility engine, consent manager, compliance tracker) reuse the same validated data-access and logging path, instead of each service re-implementing its own compliance handling and risking inconsistent record-keeping.

Security Advantage

Because every business-layer component must go through the Data & Security Layer to read or write trial data, cryptographic audit-trail generation required for 21 CFR Part 11 is enforced at a single choke point rather than left to individual services. This makes it far harder for any component to bypass audit logging or silently alter eligibility, consent, or compliance records.

Performance Benefit

Because protocol criteria and candidate records are served from one well-defined data layer shared by all business-layer services, frequently accessed inclusion/exclusion criteria can be cached at that layer. This reduces redundant database round-trips when the Eligibility Screening Engine is evaluating many candidate submissions concurrently during an active recruitment drive.